

Enterprise Hybrid Cloud Networking on AWS

Project Overview

This project demonstrates the implementation of an enterprise-grade hybrid cloud networking architecture using AWS Transit Gateway, Site-to-Site VPN, Client VPN, and a Disaster Recovery environment.

The architecture simulates a real-world organization that operates both on-premises infrastructure and AWS cloud resources while maintaining secure connectivity between all environments.

Architecture

Production Environment

CIDR: 10.1.0.0/16

Contains:

- Public Subnet
- Private Subnet
- Bastion Host
- Application Servers

Disaster Recovery Environment

CIDR: 10.2.0.0/16

Contains:

- Public Subnet
- Private Subnet
- Recovery Servers

On-Premises Environment

CIDR: 172.16.0.0/16

Contains:

- Ubuntu Router
- Internal Application Servers

Connectivity Components

- AWS Transit Gateway
 - VPN Attachment
 - Static Site-to-Site VPN
 - AWS Client VPN Endpoint
 - Route Tables
 - Security Groups
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Technologies Used

- AWS VPC
 - AWS Transit Gateway
 - AWS Site-to-Site VPN
 - AWS Client VPN
 - Libreswan
 - Ubuntu Server
 - Linux Networking
 - Route Tables
 - Security Groups
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Project Objectives

1. Implement secure hybrid connectivity between AWS and on-premises infrastructure.
 2. Design a scalable hub-and-spoke architecture using Transit Gateway.
 3. Build a disaster recovery environment.
 4. Provide secure remote user access using AWS Client VPN.
 5. Validate routing and connectivity across all environments.
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Connectivity Validation

Successfully Tested:

- ✓ Client VPN → Production VPC
- ✓ Client VPN → DR VPC
- ✓ Client VPN → On-Premises Network
- ✓ Production VPC → DR VPC

- ✓ Production VPC → On-Premises Network
 - ✓ DR VPC → On-Premises Network
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Key Learning Outcomes

- Enterprise network architecture design
 - Hybrid cloud networking
 - VPN deployment and troubleshooting
 - AWS Transit Gateway implementation
 - Disaster recovery networking
 - Remote access solutions
 - Linux router configuration
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Future Improvements

- Dynamic Routing (BGP)
- Multi-Region Disaster Recovery
- Direct Connect Integration
- Terraform Automation
- Monitoring with CloudWatch
- High Availability VPN Routers